

Alex Bauer

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WORK EXPERIENCE

Rochester Telemessaging Center

Rochester, MI

Lead Agent / Technical Support Specialist

06/2025 – Current

- Serves as primary contact for ~1,000 client companies across financial services, healthcare, medical technology, and consumer services; owns the after-hours interface for critical communications.
- Operates Abbott and Sorin pacemaker/ICD rep paging systems, resolving urgent, high-pressure escalations with discretion and technical acumen.
- Handles sensitive, time-critical communications for financial advisors, wealth managers, and estate attorneys, often speaking directly with executives and key decision-makers on clients' behalf.
- Leads the onboarding process for new associates, including technology setup and team integration.
- Promoted from Operator to Lead Agent after 11 months; supervises the floor one shift per week, advises agents on complex situations, and coordinates scheduling and coverage across the client portfolio.

Celtica French Bakery

New Orleans, LA

POS Implementation Specialist

10/2022 – 11/2024

- Led full-cycle deployment of an integrated point-of-sale system, automating data tracking and streamlining daily operations. Built automated reporting that cut manual data entry errors by over 90%, improved data integrity, and corrected tip distribution discrepancies.
- Coordinated intake, production, and delivery for hundreds of specialty King Cake orders during Mardi Gras, the bakery's highest-volume season.

BYOBeez

Appleton, WI

Assistant Researcher

09/2017 – 01/2020

- Developed machine learning models in Python and R that improved forecast accuracy by 25%.
- Created executive-level dashboards translating regression and time series analysis into actionable insights.

PROJECTS

Seraphone: Scam Call Guard

<https://seraphone.ai>

- Multi-tenant voice AI answering service in production: React SPA, FastAPI API, real-time telephony engine on Twilio and the OpenAI Realtime API, and an MCP tool server, deployed on GCP Cloud Run and Vercel with Postgres/pgvector, Redis streaming, and Stripe billing (launched 2025).
- Built the real-time governance engine (compliance guardrails, spam and hostile caller handling) and post-call pipeline (PII redaction, summaries, embeddings, caller knowledge graph); ~6,000 automated tests gate CI. Live demos: (248) 846-1901 and (248) 710-0164.

Atlas of Machine Understanding

<https://ml.aebauer.dev>

- Interactive atlas of 12,000 AI and philosophy papers: Vertex AI embeddings, UMAP projection, and RAPTOR clustering drawn as an engraved deck.gl map by an offline GPU terrain pipeline, with a streaming research agent and a GPU cross-encoder reranker with token-occlusion attribution (2026).

RAPTOR

<https://kg.aebauer.dev>

- Code knowledge graph implementing RAPTOR: Tree-sitter chunking, pgvector retrieval, and GMM-clustered LLM summary layers, with GitHub webhook updates that re-summarize only changed nodes; ships as a Cloud Run MCP server grounding coding agents in cross-repo architecture (2024).

TECHNICAL SKILLS

Python, TypeScript, FastAPI, React, SQL, PostgreSQL, pgvector, Redis, Twilio, OpenAI Realtime API, Vertex AI, RAG, vector embeddings, MCP servers, Tree-sitter, GCP Cloud Run, Stripe, pytest, CI/CD, time series forecasting

EDUCATION

Lawrence University

Appleton, WI

Bachelor of Arts in Philosophy and Biology with a Concentration in Machine Learning

2016 – 2020